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Editor-in-Chief
Information Systems Research
INFORMS

Re: Submission of the manuscript "When Should the Firm Take the Task Back? Error Dynamics and the Economics of Monitoring Autonomous AI Agents"

Dear Editor-in-Chief,

I am pleased to submit the manuscript "When Should the Firm Take the Task Back? Error Dynamics and the Economics of Monitoring Autonomous AI Agents" for consideration as a research article in Information Systems Research. The manuscript is original, has not been published, and is not under consideration at any other journal.

Firms are beginning to delegate multistep work, such as customer service and software maintenance, to autonomous AI agents while keeping a person who can take the task back. Every such deployment needs a rule that says when control should be withdrawn, and today these rules are set largely by intuition. The manuscript asks when a firm should take a task back from an agent and what its monitoring system should track to make that decision well.

The paper combines an economic model with large-scale evidence. The model treats delegation as an optimal stopping problem and distinguishes three reasons why errors matter: they may trigger further errors (contagion), tasks may differ in difficulty (heterogeneity), and an error may derail a task so that its output is wrong (derailment). Each channel calls for a different monitoring statistic, and under derailment the optimal rule is a weighted count of errors with a closed-form threshold. The value of monitoring is greatest for tasks at the margin of delegation. The evidence comes from 80,036 public trajectories of a software-engineering agent and from two randomized fault-injection experiments, in customer service and in the repair of real software bugs, with five agents, including two recent commercial models, and 11,352 runs. Most of the error clustering visible in observational data reflects heterogeneity rather than contagion. Transient failures rarely harm recent models, but errors that misreport the state of the environment derail them, lowering task success by up to 30 percentage points. What an error says matters more than how many errors occur.

I believe the manuscript fits Information Systems Research for three reasons. First, it extends research on human–AI delegation, including work published in the journal, from the decision to delegate to the decision to revoke a delegation, which becomes central once agents act over many steps. Second, it links the choice of what to monitor to IS research on control and to the economics of information, and it connects agent oversight to the field's long-standing concern with information quality: capable agents act on what their tools tell them, so the accuracy of enterprise data becomes part of AI governance. Third, it offers managers design rules that can be

implemented with existing escalation mechanisms and calibrated with a small experiment on their own tasks.

All data used are public or were generated by the author. The code, run-level results, and analysis scripts will be made available in a public replication package. The author used a generative AI assistant to help write analysis code, run the experiments, and edit the text, as disclosed on the title page; the author designed the study, verified all analyses, and takes full responsibility for the content. The author has no conflicts of interest to declare and received no external funding for this research.

Thank you for considering the manuscript. I look forward to hearing from you.

Sincerely,

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